

University Master's Degree in Machine Learning

Assignment: Image Classification with Neural Networks

The main objective of this assignment is to **analyze and compare the performance of different neural network models on an image classification dataset**. The dataset must be chosen by the students (it should not be excessively complex and must be different from those used in class). The dataset choice is only a necessary step, as the real focus lies in the analysis of the models and the conclusions that can be drawn from their performance.

Once the dataset has been selected, students will:

- Carry out a **comprehensive dataset analysis** (number of images, number of classes, class distribution, image dimensions, etc.), which will serve as context for the experiments.
- Implement and evaluate the models studied in class, from MLPs to CNNs with different configurations.
- Compare the results across models, highlighting the improvements achieved when increasing complexity, and provide a critical discussion of the findings.
- Optionally enrich the experiments with techniques covered in class, such as regularization methods, different optimizers, etc.

We expect each group to **find a way to differentiate themselves** from the others in some aspect of the work. However, this differentiation should **not** come from adding methods or techniques not covered in class. Instead, it should come from the **quality of the analysis, the originality of the insights, and the strength of the conclusions**.

Deliverables

A written report in **PDF format**, with a maximum length of approximately **10 pages**, including:

- A detailed dataset analysis
- Description of the experiments conducted (with parameters and design choices)
- Results obtained
- Critical analysis and conclusions

An **oral presentation** (10–15 minutes, followed by questions) in which each group summarizes their work, emphasizing what they consider most relevant.

The assignment should preferably be done in **pairs**, although individual submissions may be allowed upon prior approval from the instructor.

Deadline for submission of the report and presentation: Thursday, **October 8**, at 23:59.

Oral presentations: Friday, **October 9**, starting at 10:00.

Evaluation Criteria

Your work will be evaluated mainly by the oral presentation (it should be clear, engaging, well-prepared and we expect solid answers to questions). Besides, we will consider the following aspects:

- **Dataset analysis:** completeness and level of detail; the analysis itself should allow meaningful conclusions about the dataset.
- **Richness and variety of implemented models:** models must make sense for the problem, be properly designed, and show diversity.
- **Experimental design:** experiments must be clearly described, including all parameters necessary to ensure reproducibility.
- **Analysis of results and conclusions:** strong and original analysis, with conclusions fully supported by the experimental evidence.
- **Report quality:** clear, well-structured, professional report; effective use of visuals and tables.